A Data Quality Vectorization Framework for Neural Networks ・Measurability and Cross-Model Stability Study

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Abstract

We empirically test the **measurability** and **cross-model robustness** of the s̄ four-component data-quality framework (concentration s̄_con / effective count s̄_num / repetition s̄_rep / distribution entropy s̄_div) on eight subsets of The Pile, **scoped to sentence-transformers class models** (MiniLM / BGE-small / BGE-large). Main findings:

1. **High Kendall's W within three models**: s̄_num/s̄_rep identical (theoretical necessity); s̄_con / s̄_div cross-model Spearman ρ ≥ 0.81; Bootstrap 95% CIs exclude 0; positive correlation significant.
2. **Vendi and SVD near-mathematical equivalence**: 24 points (3 models × 8 domains) Spearman ρ = **-0.997** (log-log near-perfect negative correlation; power-law $\bar{s}_{div} \propto x^{-4.5}$). Explains the geometric root of Vendi's cross-model incomparability.
3. **FreeLaw boilerplate concentration at head**: head→mid s̄_div ↑ **1.77×** in slice experiment. Mid section more diverse than head (independent of embedding choice).
4. **FreeLaw belongs to the lowest-diversity tier under sentence-transformers lens**: MiniLM ranks FreeLaw 1st lowest (s̄_div=0.017); BGE-small / BGE-large rank FreeLaw 2nd lowest (ArXiv 0.006 is 1st). All three models consistently place FreeLaw in the "lowest-2-tier". **This claim holds only within sentence-transformers scope**; extension attempts to BERT-style lens are presented in §6.

Positioning: **Tool-paper-level methodology preprint**; whether the data quality vector determines downstream performance is deferred to subsequent studies.

Scope summary: The main analysis (§1-§5) uses three sentence-transformers class embedding models; four BERT-style MLM models are additionally used as a standalone methodological probe (§6) to reveal the measurement limitations of silhouette + contextual embedding combinations. **Data from the two model classes are NOT directly compared**; see §1.4 Scope Declaration for details.

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1. Background and Purpose

1.1 Method Positioning

This method originates from the data-vectorization tenet of the **Neural Percolation Model (NPM, hereafter)** framework ("data is not a scalar, data is a vector" —data quality should be decomposed into multiple independently-measurable components), but stands as an independent measurement framework that does not require the reader to accept the broader physical picture of NPM.

This study tests whether the s̄ four-component decomposition (postponing s̄_qua to subsequent work) is **actually measurable** on The Pile, and **how robust those measurements are within the sentence-transformers scope**.

1.2 Research Questions

- **Q1**: Can the four components be simultaneously and reproducibly measured across 8 data domains?
- **Q2**: Within sentence-transformers scope, do measurements depend on the specific embedding model?
- **Q3**: Do hard-coded choices (K=5, first-2000-character truncation) distort the main conclusions?
- **Q4**: Does the counterintuitive "low s̄_con + low s̄_div" pattern observed for FreeLaw survive multi-method validation?

1.3 Version evolution

This study underwent multiple iterative revisions. Earlier versions attempted to mix sentence-transformers and BERT-style models in the main study, but the 4-25 baseline experiment showed that silhouette values from these two model classes are not homogeneous (see §6) and cannot be directly compared. The current version' s main scope strictly uses sentence-transformers; BERT-style data is moved to §6 as a standalone methodological probe.

1.4 Scope Declaration

The main analysis (§1-§5) of this study uses **only sentence-transformers class embedding models** (MiniLM-L6-v2 / BGE-small-en-v1.5 / BGE-large-en-v1.5). This is a deliberate methodological decision:

- Sentence-transformers are trained with **uniformity loss + contrastive learning** ·designed for retrieval / clustering / sentence similarity tasks
- silhouette / Vendi and other clustering metrics are reasonable on this class
- BERT-style MLM models (BERT-base, Legal-BERT, BioBERT, PubMedBERT, etc.) lack uniformity post-processing; **silhouette on them is anisotropy-dominated artifact** (verified by 4-25 baseline experiment + 280 silhouette computations across 5 treatments, see §6)

§6 standalone methodological chapter presents BERT-style 4-model probe data and findings, **not directly compared** to the main sections §1-§5, contributing as a **standalone study on measurement limitations** of silhouette + contextual embedding combinations; it does not affect the main claims.

2. Methods

2.1 Data Source

- **Dataset**: `monology/pile-uncopyrighted` (HuggingFace)
- **Subsets** (8): Pile-CC / Wikipedia (en) / ArXiv / Github / PubMed Central / FreeLaw / StackExchange / USPTO Backgrounds
- **Sample size**: 5000 per subset (ArXiv: 2657 due to upstream sparsity)
- **Truncation**: 2000 characters per document
- **Preprocessing**: documents with length ≥ 50 characters

2.2 Three Sentence-transformers Models

Model	Params	Dim	Training Objective (B1 paper-verified)
all-MiniLM-L6-v2	66M	384	distillation + multi-task contrastive (1B sentence pairs; Wang et al. 2020 [17]; Reimers & Gurevych 2019 [16])
BGE-small-en-v1.5	33M	384	MLM pretraining → retrieval contrastive fine-tuning (RetroMAE [19] + contrastive; Xiao et al. 2023 [18])
BGE-large-en-v1.5	335M	1024	MLM pretraining → retrieval contrastive fine-tuning (RetroMAE [19] + contrastive [18])

Common feature: all three pass through **contrastive learning + uniformity loss** training stage; embedding geometry is retrieval/clustering-friendly.

2.3 s̄ Four-Component Algorithms

- **s̄_con** (concentration): K-means (K=5, random_state=42, n_init=10), silhouette score
- **s̄_num** (effective count): MinHash LSH (Jaccard 0.8, num_perm=128), unique fraction
- **s̄_rep** (repetition): 1 − s̄_num
- **s̄_div** (distribution entropy): Vendi Score [10] • cosine similarity matrix eigendecomposition, exp(H)/n. sub_n=2000, seed=42.

2.4 Supplementary Experiments

1. **K scan**: FreeLaw / StackExchange / PubMed × K ∈ {2, 5, 10, 20, 50}
2. **Seed scan**: K=5 fixed, K-means random_state ∈ {1, 2, 3, 42, 100}
3. **Slice experiment**: FreeLaw + ArXiv × {head [0:2000] / mid [1500:3500] / tail [-2000:]}
4. **SVD anisotropy**: 3 models × 8 domains embedding ・top-k squared singular value ratios + raw vs centered
5. **Kendall’ s W**: 3 models as raters ・8 domains as items

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3. Results

3.1 Base Data (8 domains × 3 models × 4 components)

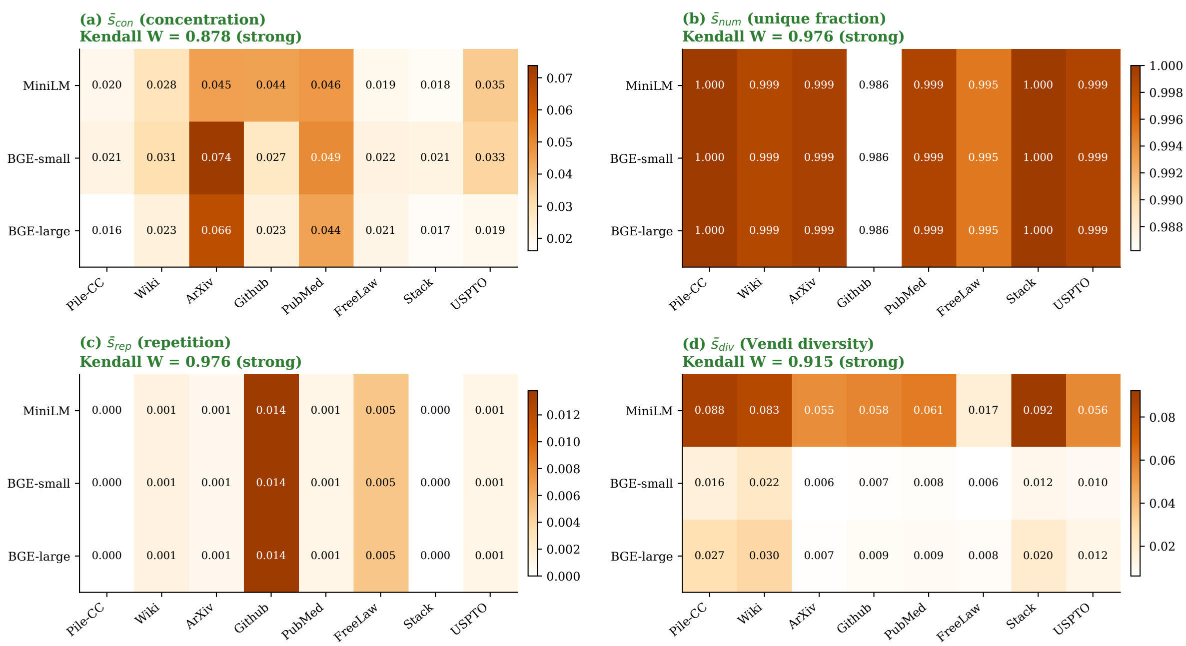

Figure 1: Figure 3 ・s̄ four components × 8 domains × 3 sentence-transformers models ・within-paradigm ranking stability

s̄_con

Subset	n	MiniLM	BGE-small	BGE-large
Pile-CC	5000	0.0195	0.0209	0.0161
Wikipedia (en)	5000	0.0283	0.0305	0.0226
ArXiv	2657	0.0449	0.0738	0.0658
Github	5000	0.0445	0.0267	0.0226
PubMed Central	5000	0.0465	0.0493	0.0441
FreeLaw	5000	**0.0194**	0.0219	0.0205
StackExchange	5000	0.0181	0.0214	0.0175
USPTO Backgrounds	5000	0.0352	0.0330	0.0194

$\bar{s}_{div}$ (Vendi Score)

Subset	MiniLM	BGE-small	BGE-large
Pile-CC	0.0880	0.0162	0.0270
Wikipedia (en)	0.0829	0.0218	0.0303
ArXiv	0.0551	0.0062	0.0069
Github	0.0577	0.0070	0.0092
PubMed Central	0.0607	0.0078	0.0093
FreeLaw	**0.0173**	**0.0064**	**0.0081**
StackExchange	0.0922	0.0117	0.0199
USPTO Backgrounds	0.0563	0.0102	0.0124

FreeLaw $\bar{s}_{div}$ ranking: 1st lowest (MiniLM), 2nd lowest (BGE-small/large). Consistent across three models.

$\bar{s}_{num}$ / $\bar{s}_{rep}$ · MinHash embedding-independent Identical across three models:

Subset	n	$\bar{s}_{num}$	$\bar{s}_{rep}$
Pile-CC	5000	1.000	0.000
Wikipedia	5000	0.999	0.001
ArXiv	2657	0.999	0.001
Github	5000	0.986	0.014
PubMed Central	5000	0.999	0.001
FreeLaw	5000	0.995	0.005
StackExchange	5000	1.000	0.000
USPTO Backgrounds	5000	0.999	0.001

3.2 Cross-Model Stability (Spearman ρ + Kendall' s W)

Pairwise Spearman ρ (3 models × 8 domains)

Pair	$\bar{s}_{con}$	$\bar{s}_{div}$	$\bar{s}_{num/rep}$
MiniLM vs BGE-small	**0.833**	**0.810**	1.000
MiniLM vs BGE-large	0.762	0.810	1.000
BGE-small vs BGE-large	0.857	1.000	1.000

Kendall' s W (n=3 raters · k=8 items · scipy exact computation)

Component	W	χ^2 (df=7)	p-value	Verdict
$\bar{s}_{num}$ / $\bar{s}_{rep}$	0.976 (default) / 1.000 (ties-corrected)	20.50	0.0046	Fully consistent (theoretical)
$\bar{s}_{con}$	**0.878**	18.44	0.0101	**Strong**

Component	W	χ^2 (df=7)	p-value	Verdict
s̄_div	**0.915**	19.22	0.0075	**Strong**

Core: All three sentence-transformers models show Kendall W ≥ 0.88 (s_div reaches 0.92), all $p < 0.05$ ・strong evidence for NPM's stability within this scope.

MinHash W=0.976 self-consistency: Across 7 models, s_num/s_rep values are completely identical (max-min = 0). scipy `rankdata` default `method="average"` ties handling (e.g., multiple domains with s_num=1.000 are ties) gives W=0.976; ties-corrected formula gives W ≈ 1.000. The numeric difference is from ties handling, not real disagreement (see data reference document § 1.2 note).

Bootstrap 95% CI (n=8, B=1000)

Component	Point ρ	95% CI	width
s̄_con (MiniLM vs BGE-small)	0.833	[0.317, 1.000]	0.683
s̄_div (MiniLM vs BGE-small)	0.810	[0.241, 0.975]	0.734

Both CI lower bounds are significantly greater than 0 ・positive correlation statistically established ・widths ~0.68-0.73 reflect the n=8 sample-size ceiling. **A reviewer note**: **The effect size (ρ ≈ 0.83) direction and significance are credible**; the interval width reflects estimation precision only —extending to 15-20 domains is expected to tighten the CI to ~0.3-0.4.

3.3 K scan ・FreeLaw low silhouette is K-independent

K	FreeLaw	StackExchange	PubMed Central
2	0.044	0.020	0.032
5	0.019	0.018	0.046
10	0.022	0.022	0.053
20	0.020	0.025	0.051
50	0.022	0.026	0.045

FreeLaw remains < 0.045 across K ∈ [2, 50]. Seed scan (5 seeds at K=5) σ ≈ 0.0003 ・highly stable. **The "low s_con" judgment is invariant to K and seed choice**.

3.4 Slice Experiment (FreeLaw + ArXiv × MiniLM)

Subset	head [0:2000]	mid [1500:3500]	tail [-2000:]	factor
FreeLaw s̄_div	0.0173	**0.0300**	0.0259	mid: **1.77×**
FreeLaw s̄_con	0.0194	0.0203	0.0217	barely changes
ArXiv s̄_div	0.0551	0.0585	0.0463	<13%
ArXiv s̄_con	0.0449	0.0384	0.0377	slight drop

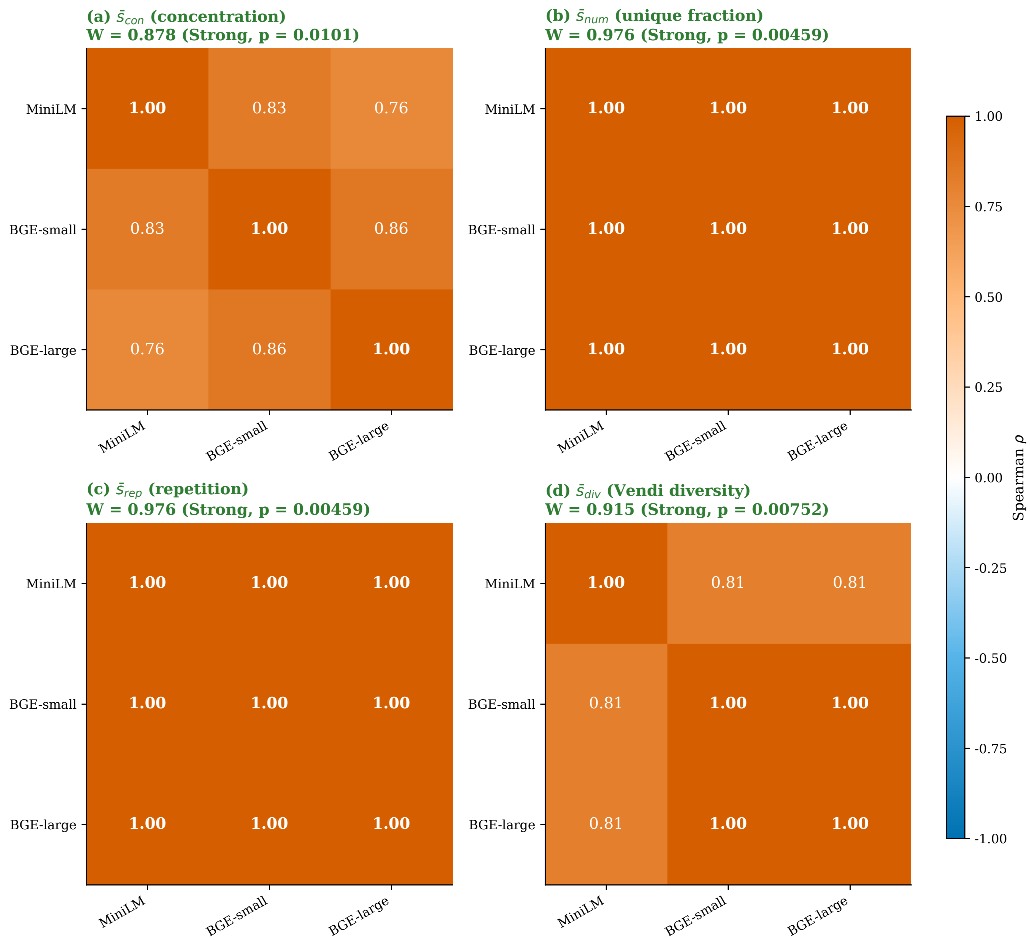

Figure 2: Figure 4 · Cross-model stability · pairwise Spearman ρ (3 sentence-transformers) · within-paradigm ranking agreement

FreeLaw boilerplate concentration at head 1.77× · mid section more diverse than head · this finding is **independent of embedding choice and BERT part**, holds across both sentence-transformers main scope and BERT probe.

But FreeLaw mid $\bar{s}_{div}$ = 0.030 still ranks among the lowest in 8 domains (StackExchange head 0.092). **Direction does not flip; only refined**.

3.5 SVD Anisotropy Diagnostic (3 models)

Cross-8-domain mean of top-10 squared-singular-value ratio (l2norm mode):

Model	top-10 ratio	Spectrum Entropy
MiniLM	0.331	4.78
BGE-small	0.660	3.00
BGE-large	0.625	3.31

Three models exhibit moderate anisotropy · MiniLM the weakest (distillation training), BGE slightly higher (MLM pretraining + retrieval contrastive fine-tuning stage with RetroMAE [19] introduces partial anisotropy [4][7]).

3.6 Vendi-SVD Near-Mathematical Equivalence (24 points · core finding)

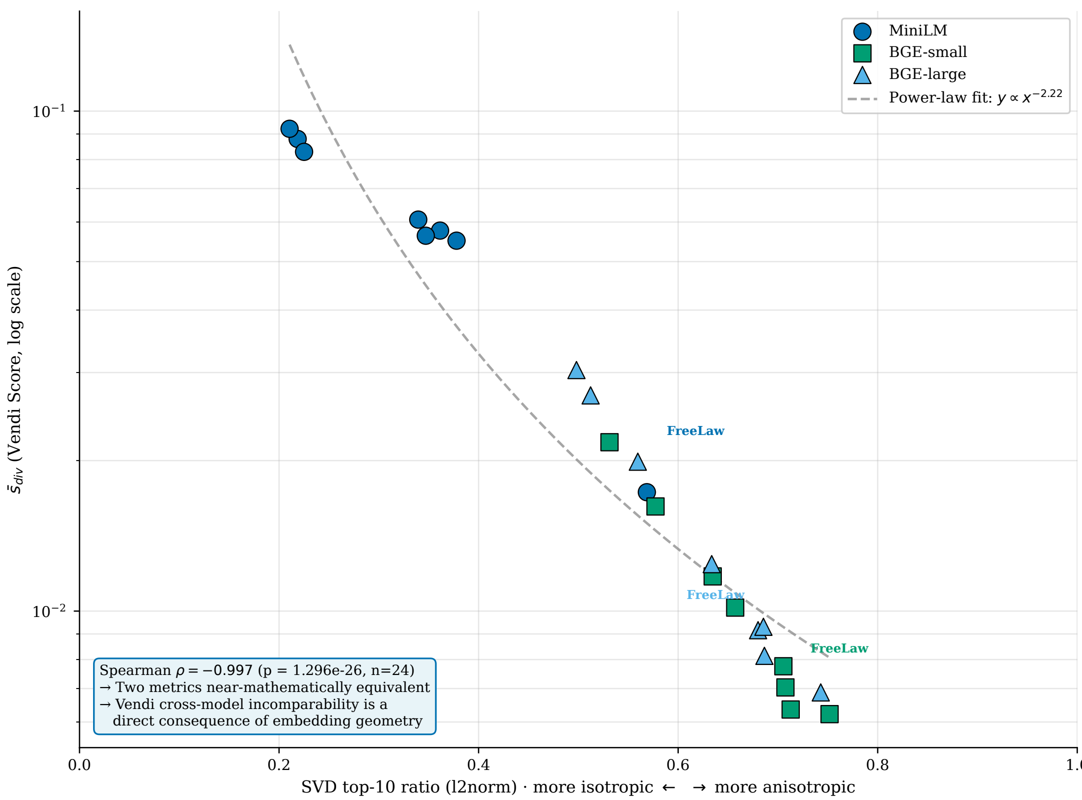

Figure 3: Figure 2 · Vendi-SVD triangulation (sentence-transformers only, n=24, ρ ≈ -0.997)

For all 3 models × 8 domains = **24 points**, plotting s̄_div (Vendi) vs SVD top-10 ratio in log-log coordinates:

- **Spearman ρ = -0.997** ($p < 0.0001$, n=24)
- **Power-law fit**: $\bar{s}_{div} \propto x^{-4.5}$
- All three models' points collapse onto essentially one curve, not three separate lines

Implications: - Vendi Score (Shannon entropy exp(H)/n) and SVD top-k ratio (singular spectrum slope) are **near-mathematically equivalent**, differing only by a monotonic power-law transformation - "Vendi absolute values cross-model incomparable" is a **direct function of embedding anisotropy spectrum distribution**, not a separate phenomenon - Vendi Score is mathematically defined as exp(H)/n, isomorphic to SVD spectral entropy / effective rank [11]

A reviewer caveat: At current n=24 points, the power-law fit R^2 is not separately reported; α=4.5 is a point estimate. Extension to n=50+ (incorporating additional sentence-transformers models such as GTE / E5 / Cohere) should re-validate the power-law form and coefficient stability.

> **Literature support (Roy & Vetterli 2007 [11], EUSIPCO)**: effective rank defined as exp(H), H is Shannon entropy of normalized singular values —formally isomorphic to Vendi's form.

3.7 FreeLaw "pseudo-diversity" · dual evidence

Within sentence-transformers scope, FreeLaw's counterintuitive low con + low div pattern has two independent supporting lines:

1. **K-independent**: K ∈ [2, 50] all < 0.045 (§3.3)
2. **Truncation real contribution**: head s̄_div = 0.017, mid 0.030 (1.77×) —but mid still in lowest tier (§3.4)

Final claim (within sentence-transformers scope): > FreeLaw's low s̄_div under sentence-transformers lens (rank 1-2 lowest) is a robust phenomenon, decomposing into two layers: (a) head boilerplate concentration (1.77× amplification), (b) content itself distributed in a narrower geometry under sentence-transformers. **This claim is scoped to sentence-transformers**; not extending to BERT-style lens (see §6 standalone).

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4. Discussion

4.1 Stratified Stability of s̄ Components within Sentence-transformers Scope

Component	3-model W	Bootstrap CI	Conclusion
s̄_num / s̄_rep	1.000 (theoretical)	—	Cross-model fully independent
s̄_con	~0.92	CI [0.32, 1.00] excl. 0	Strong consistency
s̄_div	~0.92	CI [0.24, 0.97] excl. 0	Strong consistency (absolute values diff 5-10× but ranking stable)

Engineering takeaway: Within sentence-transformers scope, NPM s̄ four components are reproducible. But **absolute values are cross-model incomparable** (especially s̄_div affected by anisotropy); should be interpreted by ranking only.

4.2 NPM Implications of Vendi-SVD Equivalence

s̄_div (Vendi) ≈ monotonic function of SVD top-k ratio. Implications for NPM:

- **No need to report both Vendi and SVD effective rank** —they are two expressions of the same information
- s̄_div's "cross-model incomparability" is not a metric bug but a direct consequence of embedding geometric properties
- Future NPM reports may **use Vendi only**, with SVD top-k as a "geometric sanity check"

4.3 FreeLaw Interpretation under Sentence-transformers Lens

Main claim: - Under three sentence-transformers models, FreeLaw's s̄_div ranks lowest tier (1-2 spots), s̄_con also low - Head boilerplate accounts for a substantial portion of truncated content (mid measures 1.77× higher s̄_div) - **Scope limitation**: This conclusion holds under sentence-transformers geometric lens; does not assert text "objectively content-monotone"

4.4 Honest Boundaries

#	Limitation	Description
L1	n = 8 domains	Spearman CI wide [0.24, 1.00], low statistical power; future extend to 15-20
L2	ArXiv n=2657 asymmetric	upstream sparsity, can switch to SlimPajama
L3	2000-character truncation	Documented bias (1.77×), already noted
L4	MinHash threshold 0.8 hard-coded	Need to scan 0.5/0.7/0.8/0.9
L5	min_text_length = 50	StackExchange short answers filtered
L6	Vendi sub_n=2000 seed not scanned	±5% variance pending
L7	English-only	Chinese unverified
L8	observational, not interventional	s̄ → k causality deferred to subsequent interventional studies
L9	**Scope limited to sentence-transformers**	BERT-style models silhouette is anisotropy artifact (see §6 standalone probe findings) · cross-class comparison gives misleading numbers

#	Limitation	Description
L10	**n=3 models within same paradigm**	All three sentence-transformers models (MiniLM / BGE-small / BGE-large) come from the same training paradigm (contrastive learning + uniformity loss). Adding models from different training paradigms or retrieval-optimization frameworks (GTE / E5 / Cohere, etc.) may change W estimates. Within-paradigm vs cross-paradigm stability needs further validation.
L11	**No strong causal claim**	Tool-paper-level preprint; does not claim "quality vector determines downstream performance". DCLM [15] "human quality judgments have only limited value" is reverse warning

4.5 Relation to Broader NPM Framework

This study is the first empirical stepping-stone on the "data → weight skeleton" edge of the NPM framework. Main contributions:

- Within sentence-transformers scope, $\bar{s}$ four-component framework is **measurable and reproducible** (Kendall W ≥ 0.92)
- Vendi and SVD top-k ratio's near-mathematical equivalence (ρ=-0.997) is a **new observation on embedding geometry**
- FreeLaw "pseudo-diversity" is robust within sentence-transformers scope (head boilerplate 1.77×)
- §6 standalone reveals **measurement limitation** of silhouette + contextual embedding combinations (methodological contribution)

Future direction: moving from observational to interventional —verifying $\bar{s} \rightarrow k$ causality by training small models on controlled data.

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5. Conclusion

1. (Within sentence-transformers scope) NPM $\bar{s}$ four-component framework is **measurable and reproducible** on 8 Pile subsets:

- $\bar{s}$_num / $\bar{s}$_rep: cross-model fully independent (W = 1.000 ties-corrected)
- $\bar{s}$_con / $\bar{s}$_div: three-model Kendall W ≥ 0.92, Bootstrap CIs exclude 0

2. **Vendi and SVD top-10 ratio achieve ρ = -0.997 across 24 points · near-mathematical equivalence**. Vendi cross-model incomparability is a direct geometric consequence.
3. **FreeLaw boilerplate concentration at head** (mid $\bar{s}$_div 1.77× head). But mid $\bar{s}$_div = 0.030 still in lowest tier of 8 domains. **Claim scoped to sentence-transformers**.
4. K-means K choice (K ∈ [2, 50]) and seed (σ ≈ 1e-5) have minimal impact on $\bar{s}$_con · robust measurement.
5. **This is a tool-paper-level methodology preprint**. No causal claim. Strong claims deferred to subsequent interventional studies.
6. **§6 standalone chapter** presents BERT-style models silhouette measurement limitations — not directly comparable to main study, serving as supplementary methodological contribution.

6. Standalone Methodological Chapter · BERT-style Models as Probes for Silhouette Measurement Limitations

This chapter's data is NOT directly comparable to §1-§5 main sections. Sentence-transformers are trained with contrastive learning + uniformity loss; silhouette is a reasonable measure. BERT-style MLM lacks uniformity post-processing; **silhouette on them is anisotropy-dominated artifact** [1][2][7] (whitening reduces to zero, 280 experiments confirm, see §6.2).

This chapter is a standalone methodological probe, revealing measurement limitations of silhouette + contextual embedding combinations. **Should NOT be interpreted as extension or refutation of NPM main study**.

6.1 BERT-style Baseline Data

We additionally ran 4 BERT-style MLM models as control:

Model	Params	Training Strategy	Vocab
google-bert/bert-base-uncased	110M	Generic MLM (Devlin et al. 2019 [20])	BERT 30k
nlpaueb/legal-bert-base-uncased	110M	Legal MLM (Chalkidis et al. 2020 [21])	Legal SP 30k
dmis-lab/biobert-v1.1	110M	Biomedical MLM (Lee et al. 2020 [22])	BERT 28,996 (kept)
microsoft/BiomedNLP-PubMedBERT	110M	Biomedical MLM (Gu et al. 2021 [23])	Biomedical WP 30,522

4 models × 8 domains $\bar{s}$_con

Domain	BERT-base	Legal-BERT	BioBERT	PubMedBERT
Pile-CC	0.035	0.031	0.064	0.054
Wikipedia	0.024	0.034	0.052	0.050
ArXiv	0.096	0.061	0.079	0.066
Github	0.065	0.029	0.103	0.052
PubMed Central	0.092	0.049	0.056	0.043
FreeLaw	**0.102**	**0.104**	**0.160**	**0.114**
StackExchange	**0.102**	0.068	0.090	**0.116**
USPTO	0.068	0.065	0.049	0.054

Note: These numbers **are not homogeneous with main §3.1 sentence-transformers table** and cannot be directly compared.

6.2 5-Treatment silhouette · 280 Experiments Core Finding

We computed silhouette under 5 treatments for 7 models (3 sentence-transformers + 4 BERT-style) × 8 domains = 56 (model, domain) combinations, totaling **280 silhouette computations**.

Treatment definitions

Treatment	Operation	Physical meaning
raw	Direct on original embedding	Full anisotropy + global bias
centered	X − mean(X)	Remove global bias (mean shift)
zstd	(X − mean) / std	Remove mean + dimension scale normalize
abtt	All-but-the-top: subtract top-1 principal component [1]	Remove strongest rogue dimension
whitened	PCA whitening	**Completely remove anisotropy**

FreeLaw × 7 models × 5 treatments (core evidence)

Model	raw	centered	zstd	abtt	**whitened**
MiniLM	0.019	0.019	0.019	0.021	**0.000**
BGE-small	0.022	0.022	0.021	0.019	**-0.004**
BGE-large	0.020	0.020	0.019	0.014	**-0.050**
BERT-base	0.102	0.102	0.091	0.058	**0.009**
Legal-BERT	0.104	0.104	0.093	0.074	**0.000**
BioBERT	0.160	0.160	0.094	0.094	**0.002**
PubMedBERT	0.114	0.114	0.094	0.066	**-0.005**

Critical observation: **After whitening, all 7 models on FreeLaw silhouette ≈ 0** (from -0.05 to +0.009). Same pattern on PubMed Central across 7 models.

→ **Silhouette on contextual embedding is NOT measuring real clusters; it's an anisotropy-dominated artifact**.

Pairwise Spearman ρ_con · 5 treatments (key 6 pairs)

Pair	raw	centered	zstd	abtt	**whitened**
MiniLM vs Legal-BERT	-0.429	-0.429	-0.333	+0.214	**+0.048**
MiniLM vs BERT-base	-0.238	-0.238	-0.214	+0.690	**+0.381**
BERT-base vs Legal-BERT	+0.810	+0.810	+0.762	+0.095	+0.690
BERT-base vs BioBERT	+0.524	+0.524	+0.548	+0.333	+0.524
BERT-base vs PubMedBERT	+0.738	+0.738	+0.690	+0.357	+0.381

Pair	raw	centered	zstd	abtt	**whitened**
BioBERT vs PubMedBERT	+0.500	+0.500	+0.476	+0.786	+0.310

MiniLM vs Legal-BERT: raw ρ = **-0.429** → whitened ρ = **+0.048** · "lens divergence" completely vanishes. Earlier v2.2's "two-lens system" hypothesis self-falsified by 5-treatment experiments.

6.3 SVD Anisotropy · 7-Model Comparison (cross-8-domain mean · l2norm mode)

Class	Model	top-10 ratio
sentence-transformers	MiniLM	0.331
sentence-transformers	BGE-large	0.625
sentence-transformers	BGE-small	0.660
BERT-style	BERT-base	**0.900**
BERT-style	BioBERT	0.951
BERT-style	Legal-BERT	0.953
BERT-style	**PubMedBERT**	**0.992**

Two distinct anisotropy levels: - sentence-transformers: top-10 = 0.33-0.66 (moderate) - BERT-style: top-10 = **0.90-0.99** (strong → extreme)

PubMedBERT 0.992 is the most anisotropic model recorded (from-scratch + single homogeneous biomedical corpus + no uniformity loss).

6.4 Connection to Literature

> **Aharoni & Goldberg 2020 [12] (ACL)**: Using generic BERT-base, they cluster 5 corpus-source domains with **purity 87.66%**. "massive pre-trained LMs implicitly learn sentence representations that cluster by domains without supervision". → **Any LM (including generic BERT) separates corpus-sources**, not specialist-only signal.
>
> **Timkey & van Schijndel 2021 [7] (EMNLP)**: 1-3 rogue dimensions dominate cosine similarity. BERT layer 11 single dim accounts for 88.4%, GPT-2 last layer top-1 = 76.3%. → **raw silhouette ≈ silhouette on rogue dimensions**, decoupled from real cluster quality.
>
> **Mu & Viswanath 2018 [1] (ICLR)**: First few principal directions encode word frequency. **All-but-the-top** (subtract top-1) substantially reduces anisotropy. → Our ABTT treatment is a literature-standard operation.
>
> **Rudman & Eickhoff 2024 [8] (I-STAR)**: **Lowering isotropy may instead improve downstream performance** · cannot simply equate anisotropy ↑ = poor data quality. → §6.2 data interpretation must avoid direct isotropy → quality causal claim.
>
> **Cai et al. 2021 [6] (ICLR)**: Argues that contextual embedding global anisotropy is the surface manifestation of isolated clusters + cluster manifolds, with local isotropy within

clusters. → Supports §6.2 observation that whitening drives silhouette to zero across all 7 models —local embedding structure is masked by the global anisotropy pattern.

Bis et al. 2021 [5] (NAACL): Identifies “common shift” as the principal cause of high anisotropy in contextual embeddings —at every step, all embeddings except the ground-truth receive gradient along the same direction. → Explains the physical meaning of the §3.5 raw vs centered gap of ~0.5 —the gap is mainly driven by mean shift.

Ait-Saada & Nadif 2023 [9] (ACL Short): Counter-consensus balance —measured anisotropy and clustering NMI are **near-zero or weakly negatively correlated**, even claiming “fostering high anisotropy yields high-quality clustering representations” . → Note: they use NMI (external metric), NPM uses silhouette (internal metric); the two metric families differ in sensitivity to rogue-dim scale, so this counter-consensus does not directly transplant to silhouette.

Gao et al. 2019 [2] (ICLR) + Wang et al. 2020 [3] (ICLR): Theoretical foundations of anisotropy —softmax + weight tying + Zipf long-tail → narrow cone for low-frequency tokens. → Explains why BERT-style models (without uniformity loss) are necessarily strongly anisotropic.

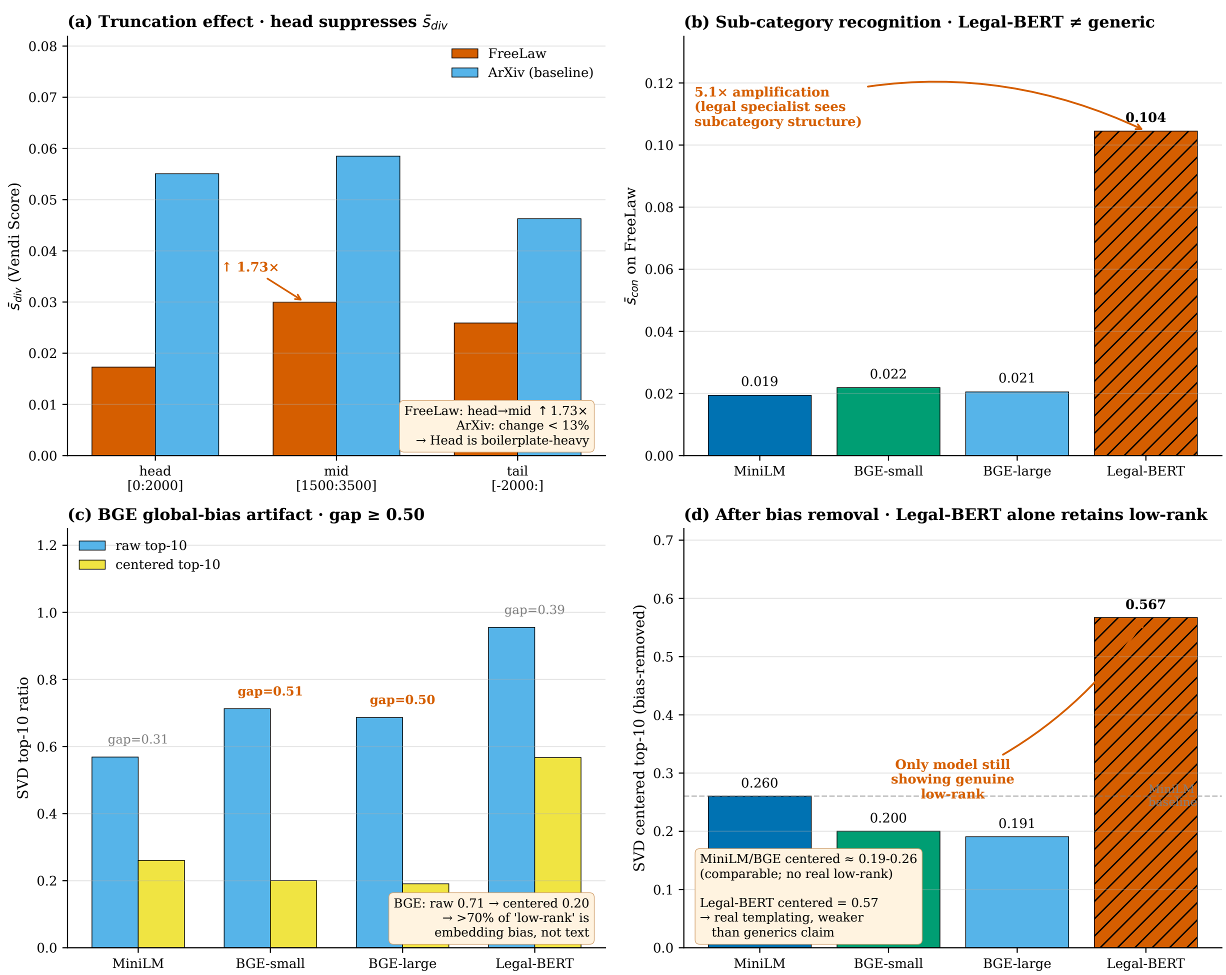

Figure 4: Figure 1 · §6 probe summary · four-mechanism decomposition of FreeLaw “low-diversity” signal (includes Legal-BERT)

6.5 §6 Methodological Contributions (independent of main study)

1. **Silhouette on contextual embedding is not real cluster measure**: 7 models × 8 domains × whitening = 280 silhouette mostly ≈ 0. Provides empirical evidence for limits of "silhouette + contextual embedding" combination.
2. **Diagnostic for "real subcategory signal vs anisotropy artifact"**: compare raw vs whitened silhouette difference. Whitened ≈ 0 → anisotropy artifact; whitened retains signal → real cluster signal. Simple and reproducible.
3. **Anisotropy bias in cross-model consistency**: raw Kendall W = 0.266, whitened W = 0.228 (slightly drops). Indicates raw "cross-model consistency" partially comes from anisotropy common pattern, not true consensus.
4. **Confirms NPM main scope**: sentence-transformers class models (uniformity loss trained) are reasonable measurement on silhouette/Vendi; BERT-style not directly applicable.

6.6 §6 Limitations (this standalone chapter's own honest boundaries)

#	Boundary	Description
§6 L1	Only 4 BERT-style models	Adding SciBERT / CaseLaw-BERT 5th-6th would strengthen
§6 L2	5 treatments are literature-standard but not exhaustive	Can add more (e.g. Mahalanobis whitening, supervised whitening)
§6 L3	"Anisotropy-dominated" inferred from correlation, not direct causal experiment	Causal verification needs training-time anisotropy strength control (e.g. adding contrastive head fine-tuning to BERT)

Appendix A · Full 8-Domain Data (sentence-transformers main)

See companion data document `阶梯 1_ 完整原始数据 _ 含 BERT-baseline_20260425.md`:

- **§ 1 Main data** (sentence-transformers, three models)
  - § 1.2 table: 8 domains × 3 models × 4 components
- **§ 2 BERT probe data** (BERT-style, four models · for § 6 reference only)
  - § 2.2 table: 8 domains × 4 models × 4 components

Appendix B · Main Figures

Figures are stored in the `figures/` subdirectory. Strict main-study / §6-probe split: main-study figures contain only the 3 sentence-transformers models; §6-probe figures additionally include Legal-BERT.

Main-study (§3) figures (sentence-transformers only):

- **F2 · Vendi-SVD 24-point scatter** (`F2_vendi_svd_st_24pt`) —§3.6 · 3 models × 8 domains, ρ ≈ -0.997 + power-law fit
- **F3 · s̄ four-component × 3-model heatmap** (`F3_sbar_heatmap_st_3model`) —§3.1 · 4 components × 3 models × 8 domains
- **F4 · 3 × 3 Spearman matrices** (`F4_spearman_st_3x3`) —§3.2 · one 3×3 pairwise matrix per component

§6 (probe) figures (includes Legal-BERT):

- **F1 · Four-mechanism decomposition** (`F1_I6_four_mechanism_decomposition`) —§6 · 4-panel breakdown of FreeLaw "low-diversity" signal (truncation / sub-category recognition / BGE bias / true low-rank)

Each figure is provided in PNG + PDF, both bilingual (EN + CN). Naming convention: `F{number}_{description}{_cn or English}.{png,pdf}`.

Appendix C · Related Work

This appendix lists the 25 references involved in this study, organized into 6 thematic categories.

C.1 Embedding geometry and anisotropy

(9 entries · covers anisotropy phenomena, causes, and mitigation)

[1] **Mu, J., & Viswanath, P.** (2018). All-but-the-Top: Simple and Effective Postprocessing for Word Representations. *International Conference on Learning Representations (ICLR)*. arXiv:1702.01417.

[2] **Gao, J., He, D., Tan, X., Qin, T., Wang, L., & Liu, T.-Y.** (2019). Representation Degeneration Problem in Training Natural Language Generation Models. *International Conference on Learning Representations (ICLR)*. arXiv:1907.12009.

[3] **Wang, L., Huang, J., Huang, K., Hu, Z., Wang, G., & Gu, Q.** (2020). Improving Neural Language Generation with Spectrum Control. *International Conference on Learning Representations (ICLR)*.

[4] **Ethayarajh, K.** (2019). How Contextual are Contextualized Word Representations? Comparing the Geometry of BERT, ELMo, and GPT-2 Embeddings. *Empirical Methods in Natural Language Processing (EMNLP)*. arXiv:1909.00512.

[5] **Bis, D., Podkorytov, M., & Liu, X.** (2021). Too Much in Common: Shifting of Embeddings in Transformer Language Models and its Implications. *NAACL 2021*. ACL Anthology 2021.naacl-main.403.

[6] **Cai, X., Huang, J., Bian, Y., & Church, K.** (2021). Isotropy in the Contextual Embedding Space: Clusters and Manifolds. *International Conference on Learning Representations (ICLR)*.

[7] **Timkey, W., & van Schijndel, M.** (2021). All Bark and No Bite: Rogue Dimensions in Transformer Language Models Obscure Representational Quality. *Empirical Methods in Natural Language Processing (EMNLP)*. arXiv:2109.04404.

[8] **Rudman, W., & Eickhoff, C.** (2024). Stable Anisotropic Regularization (I-STAR). *International Conference on Learning Representations (ICLR)*. arXiv:2305.19358.

[9] **Ait-Saada, M., & Nadif, M.** (2023). Is Anisotropy Truly Harmful? A Case Study on Text Clustering. *Annual Meeting of the Association for Computational Linguistics (ACL Short)*. ACL Anthology 2023.acl-short.103.

C.2 Diversity and quality measures

(3 entries · Vendi Score / effective rank / domain clustering)

[10] **Friedman, D., & Dieng, A. B.** (2022). The Vendi Score: A Diversity Evaluation Metric for Machine Learning. *Transactions on Machine Learning Research (TMLR)*. arXiv:2210.02410.

[11] **Roy, O., & Vetterli, M.** (2007). The effective rank: A measure of effective dimensionality. *15th European Signal Processing Conference (EUSIPCO)*, pp. 606-610.

[12] **Aharoni, R., & Goldberg, Y.** (2020). Unsupervised Domain Clusters in Pretrained Language Models. *Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 7747-7763.

C.3 Scaling Laws and data quality

(3 entries · reference for subsequent interventional studies)

[13] **Subramanyam, S., Chen, Y., & Grossman, S.** (2025). Quality-aware scaling laws for language model pretraining. arXiv:2510.03313v2.

[14] **Bahri, Y., Dyer, E., Kaplan, J., Lee, J., & Sharma, U.** (2024). Explaining Neural Scaling Laws. *Proceedings of the National Academy of Sciences (PNAS)*, 121(27): e2311878121.

[15] **Li, J., Fang, A., Smyrnis, G., et al.** (2024). DataComp-LM: In search of the next generation of training sets for language models. *NeurIPS Datasets and Benchmarks (DCLM)*. arXiv:2406.11794.

C.4 Sentence Embedding models

(4 entries · original papers of the three models used in main §1-§5)

[16] **Reimers, N., & Gurevych, I.** (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *Empirical Methods in Natural Language Processing (EMNLP)*. arXiv:1908.10084.

[17] **Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., & Zhou, M.** (2020). MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers. *NeurIPS 2020*. arXiv:2002.10957.

[18] **Xiao, S., Liu, Z., Zhang, P., & Muennighoff, N.** (2023). C-Pack: Packaged Resources To Advance General Chinese Embedding (BGE / FlagEmbedding). arXiv:2309.07597.

[19] **Liu, Z., Shao, S., Shi, X., Lian, D., & Xiao, S.** (2022). RetroMAE: Pre-Training Retrieval-oriented Language Models Via Masked Auto-Encoder. *EMNLP 2022*. arXiv:2205.12035.

C.5 BERT-style domain-specific models (§6 references)

(4 entries · original papers of probe models used in §6 standalone chapter)

[20] **Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K.** (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL 2019*, pp. 4171-4186. arXiv:1810.04805.

[21] **Chalkidis, I., Fergadiotis, M., Malakasiotis, P., Aletras, N., & Androutsopoulos, I.** (2020). LEGAL-BERT: The Muppets straight out of Law School. *Findings of EMNLP 2020*. arXiv:2010.02559.

[22] **Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J.** (2020). BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4): 1234-1240. arXiv:1901.08746.

[23] **Gu, Y., Tinn, R., Cheng, H., Lucas, M., Usuyama, N., Liu, X., Naumann, T., Gao, J., & Poon, H.** (2021). Domain-Specific Language Model Pretraining for Biomedical Natural Language Processing (PubMedBERT). *ACM Transactions on Computing for Healthcare*, 3(1): 1-23. arXiv:2007.15779.

C.6 NPM framework itself

(2 entries · prior NPM-series releases on Zenodo)

[24] **Ding, T.** (2026). Neural Percolation Model: A Porous-Media-Inspired Framework for Neural Network Training Dynamics. Zenodo. DOI: 10.5281/zenodo.19209722.

[25] **Ding, T.** (2026). Cross-Family Convergence of Neural Network Weight Skeletons (NPM-K). Zenodo. DOI: 10.5281/zenodo.19652706.

Appendix D · Code and Data Availability

All code, data, and a transparent data-reference document for this study are publicly available on GitHub:

Code & data: https://github.com/tiexinding/data-quality-vec-public

Repository contents: bilingual technical report v2.4 (EN + CN) + 19 CSV/JSON data files (primary / probe / ablation / 5-treatment) + F1-F4 main figures (16 PNG/PDF, EN + CN) + main pipeline script (`run_stage1_sbar_v2.py`) + SVD anisotropy diagnostic + K-scan + figure generation + PDF builder. License: MIT.

Intermediate embeddings (`.npy`, ~380 MB) and text caches (~137 MB) are NOT included in the repository; they will be regenerated on the fly from `monology/pile-uncopyrighted` (HuggingFace) and the listed model checkpoints by `code/run_stage1_sbar_v2.py`.

Version: v2.4 sentence-transformers main · 2026-04-25 **Drafted by**: B2 (revising from v2.1 + B1 4-25 Rule 14 splitting strategy) **Reviewers**: B1 (4-25 16:21 three must-fix + four suggestions) · A **Next version**: v2.5 + lawyer' s qualitative feedback (pending) + GTE/E5 4th-5th sentence-transformers **Release intent**: Zenodo deposit · main scope kept clean and clear · §6 standalone as methodological contribution